Project 10. Inverse problem for the heat equation [A]

Let $\Omega = (0,1)^3$, with bottom face $\Gamma_D = \{\mathbf{x} \in \partial\Omega : x_3 = 0\}$ and $\Gamma_N = \partial\Omega \setminus \Gamma_D$. Consider the heat equation:

$$\begin{cases} \dfrac{\partial u}{\partial t} - \mu \Delta u = 0 & \text{in } \Omega \times (0,T] \,, \\ u = u_D & \text{on } \Gamma_D \times (0,T] \,, \\ \mu \nabla u \cdot \mathbf{n} = 0 & \text{on } \Gamma_N \times (0,T] \,, \\ u = 0 & \text{in } \Omega \times \{0\} \,, \end{cases} \tag{11}$$

where μ is a constant taken in $[\mu_{\min}, \mu_{\max}]$ is the unknown thermal diffusivity and $u_D > 0$ is a given temperature imposed on Γ_D. Train a neural network [12] in `JAX` [4] that approximates the parametric map

$$(\mathbf{x}, t, \mu) \mapsto u(\mathbf{x}, t; \mu) \,, \qquad \mathbf{x} \in \Omega,\ t \in [0,T],\ \mu \in [\mu_{\min}, \mu_{\max}] \,.$$

Discuss the choice of the network architecture, of the training data and of the loss function.

Fix a value $\mu^* \in [\mu_{\min}, \mu_{\max}]$. A sensor located in a region $\omega \subset \bar{\Omega}$ acquires the temperature at points $\{\mathbf{x}_i\}_{i=1}^{N_s} \subset \omega$ every Δt_s seconds, i.e. at times $t_j = j\,\Delta t_s$, $j = 1, \dots, N_t$. The measurements are

$$d_{ij} = u_h(\mathbf{x}_i, t_j; \mu^*) + \eta_{ij} \,,$$

where u_h is an accurate finite element solution and η_{ij} is a random noise. Using `JAX`, solve the inverse problem of recovering μ^* from the measurements $\{d_{ij}\}$:

$$\widehat{\mu}^* = \operatorname*{argmin}_{\mu \in [\mu_{\min}, \mu_{\max}]} \sum_{i=1}^{N_s} \sum_{j=1}^{N_t} |d_{ij} - u_{\mathrm{NN}}(\mathbf{x}_i, t_j; \mu)|^2,$$

where $u_{NN}(\mathbf{x}_i, t_j; \mu)$ represents the evaluation of the trained neural network at the point $(\mathbf{x}_i, t_j; \mu)$. Describe the optimization algorithm used to solve it. Assess the accuracy and the robustness of the identification procedure for increasing levels of noise in the data.